Simulation-based Time Series Analysis:

Likelihood-free Estimation, Testing for Structure, and Prediction with Transformers using Synthetic Data

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Overview of presentation

- Introduction to time series

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- Conditional independence testing

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- Simulation-based estimation

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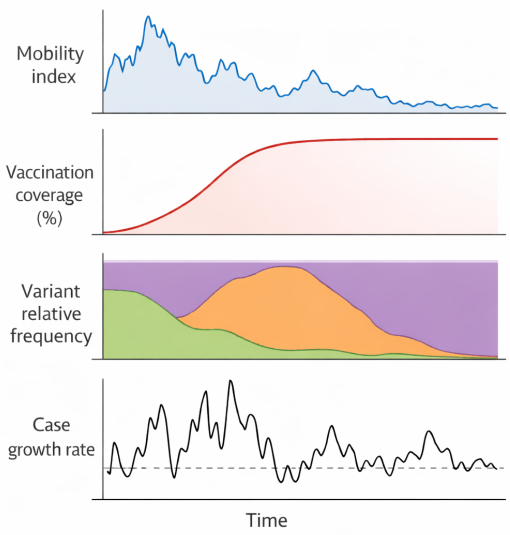
- Example: time series in epidemiology

COVID-19 time series May 2020–2022 (n=365x2=730, d=4)



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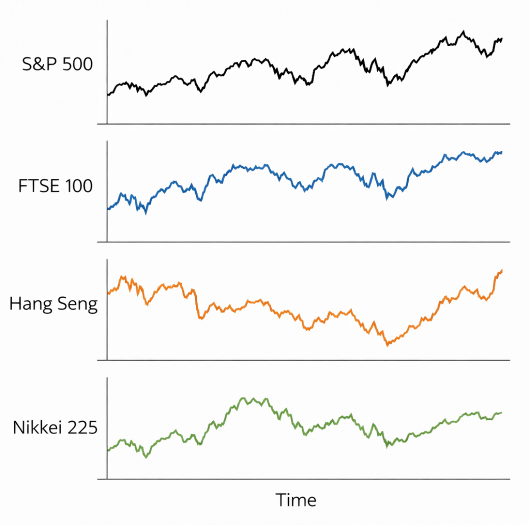
- We show a subset of the results based on BH-adjusted p-values

Closing prices of each stock index over time



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- We reject *every* null hypothesis at the significance level $\alpha = 0.05$

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- Next, we present the testing procedure

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- Calculate test statistic on time series of residual products

$$T(\hat{R}_{1:n}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \hat{R}_t \right\|_p$$

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- Reject null hypothesis at level α if $T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\mathrm{boot}}$, else retain

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- Let Θ be a parameter space, which can be infinite-dimensional
- Let $G_t^{(n)} : \mathbb{R}^\infty \times \Theta \to \mathbb{R}^d$ be some measurable function, $t, n, d \in \mathbb{N}$

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- Denote the collections by $\mathcal{P}_n = \{P_{\theta,n} : \theta \in \Theta\}$ and $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$

Key assumption: Decaying influence of noise inputs (Wu 2005)

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- There exist $\Psi, \rho > 0$ such that, for some $q > 2$ and all $j \in \mathbb{N}$, we have

$$\sup_{\theta\in\Theta}\sup_{\ell\in\mathbb{Z}}\sup_{i\in\mathbb{N}}\sup_{t\in\mathbb{N}} \left\| G_t^{(i)}(\boldsymbol{\epsilon}_\ell, \theta) - G_t^{(i)}(\tilde{\boldsymbol{\epsilon}}_{\ell,j}, \theta)\right\|_{L^q(\theta)} \leq \Psi (j \vee 1)^{-\rho}$$

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- We make this assumption for $X_{1:n}$, $Y_{1:n}$, and $Z_{1:n}$

We prove our test has uniformly asymptotic Type I error control

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Theorem (Informal)

Recall the test statistic $T(\hat{R}_{1:n})$ and estimated quantile $\hat{q}_{1-\alpha}^{\mathrm{boot}}$. We have

$$\limsup_{n \to \infty} \sup_{P \in \mathcal{P}_n^*} \mathbb{P}_P\left(T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\mathrm{boot}}\right) \leq \alpha.$$

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- For nonstationary time series, need rolling-window estimator (not discussed here)

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- Probabilistic Whitney (1936) embedding theorem due to Sauer, J. A. Yorke, and Casdagli (1991)
- For "almost every" C^1 smooth function Φ from $\mathbb{R}^p$ to $\mathbb{R}^{2p+1}$:
  1. Φ **is one-to-one** on compact subsets of $\mathbb{R}^p$
  2. Φ **has a nice derivative** on certain compact subsets of $\mathbb{R}^p$

  – Details: the $(2p+1)\times p$ Jacobian matrix of Φ has full rank p on each compact subset of a smooth manifold contained in $\mathbb{R}^p$

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- If physical dependence measure (Wu 2005) decays uniformly over Θ, we can estimate $\Phi(\theta)$, $\theta \in \Theta$, via time-averages of $2p+1$ random features

$$\frac{1}{n-m}\sum_{t=m+1}^{n} \varphi\left([X_t(\theta), \ldots, X_{t-m}(\theta)]^\top\right)$$

Example: Mean of Gaussian

- **Goal:** Estimate $\theta_0 = 0$ given $X_1^{\mathrm{obs}}, \ldots, X_n^{\mathrm{obs}} \overset{\mathrm{iid}}{\sim} \mathcal{N}(\theta_0, 1)$ with $\Theta = [-3, 3]$

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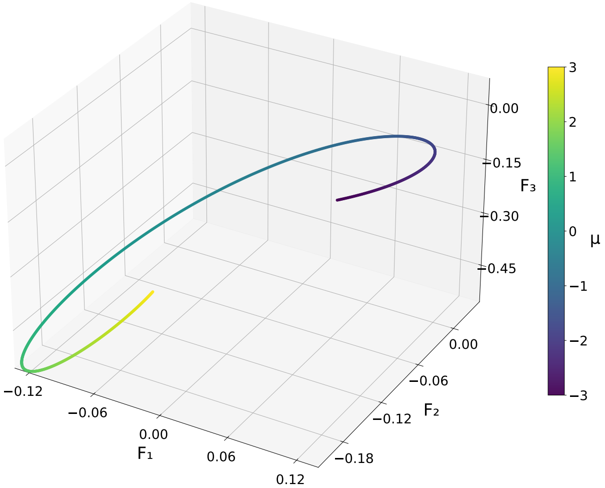
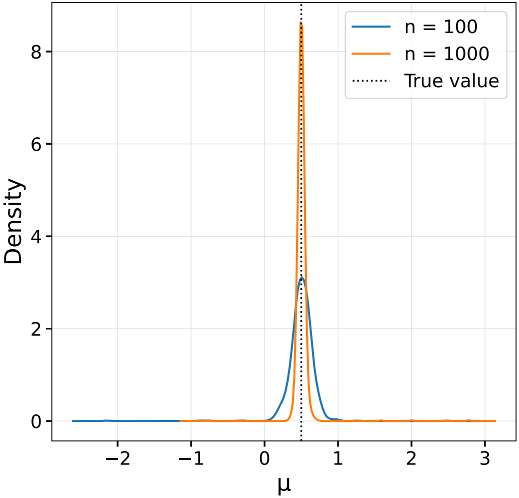
- We will compare MSE our random feature estimator $\hat{\theta}$ with $\hat{\theta}^{\text{MLE}} = \frac{1}{n} \sum_{t=1}^{n} X_t$

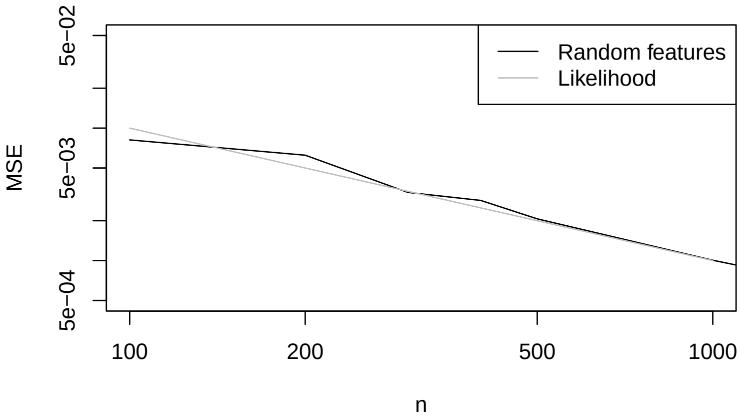
Time-average of 3 features, colored by parameter value ($n = 1000$, $s = 10$)



Density plot based on $1,000$ independent estimates



Theoretical MSE of MLE vs MSE of our estimator (100 replicates per n)



Theoretical guarantees for random feature estimators

- We prove consistency $\hat{\theta} \xrightarrow{p} \theta_0$ and asymptotic normality

$$\sqrt{n}\left(\hat{\theta} - \theta_0\right) \Rightarrow N\left(0, (\Gamma^\top \Gamma)^{-1}\Gamma^\top V \Gamma (\Gamma^\top \Gamma)^{-1}\right)$$

where Γ is the $(2p+1) \times p$ Jacobian matrix of Φ evaluated at θ_0, and V is the $(2p+1) \times (2p+1)$ asymptotic covariance matrix of $\frac{1}{n}\sum_{t=1}^{n} \varphi(X_t^{\text{obs}})$

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- Examples: ODEs with dependent observational noise, nonstationary state-space models, and structural time series models

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- We develop theory, and propose methods for smoothing and forecasting

Simulate to get pre-training data for forecasting with TSFMs

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$$Y_t = \mathrm{Se}\left(t, \theta^{Se}\right) \mathrm{Tr}\left(t, \theta^{Tr}\right) \exp\left(\kappa X_t - \frac{\kappa^2 \sigma_X^2}{2(1-\phi^2)}\right) \exp\left(\varepsilon_t^Y - \frac{\sigma_Y^2}{2}\right)$$
$$X_t = \phi X_{t-1} + \varepsilon_t^X$$

for some scaling factor $\kappa > 0$ and autoregressive coefficient $\phi \in (-1, 1)$, with $X_0 \sim N(0, \frac{\sigma_X^2}{1-\phi^2})$ and $\varepsilon_t^X \overset{\text{iid}}{\sim} N(0, \sigma_X^2)$, $\varepsilon_t^Y \overset{\text{iid}}{\sim} N(0, \sigma_Y^2)$ for some $\sigma_X, \sigma_Y > 0$

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- Trend: product of linear trend and exponential trend

Simulate to get pre-training data for forecasting with TSFMs

- For forecasting with TSFMs, Dooley et al. (2023) suggest training on simulations from *multiplicative seasonality-trend-noise decomposition models* (many $\theta \in \Theta$)
- We consider an extension with multiplicative autoregressive noise

$$Y_t = \mathrm{Se}\left(t, \theta^{Se}\right) \mathrm{Tr}\left(t, \theta^{Tr}\right) \exp\left(\kappa X_t - \frac{\kappa^2 \sigma_X^2}{2(1-\phi^2)}\right) \exp\left(\varepsilon_t^Y - \frac{\sigma_Y^2}{2}\right)$$
$$X_t = \phi X_{t-1} + \varepsilon_t^X$$

for some scaling factor $\kappa > 0$ and autoregressive coefficient $\phi \in (-1, 1)$, with $X_0 \sim N(0, \frac{\sigma_X^2}{1-\phi^2})$ and $\varepsilon_t^X \overset{\text{iid}}{\sim} N(0, \sigma_X^2)$, $\varepsilon_t^Y \overset{\text{iid}}{\sim} N(0, \sigma_Y^2)$ for some $\sigma_X, \sigma_Y > 0$

- Seasonality: linear combination of sines, cosines (weekly, monthly, yearly periods)
- Trend: product of linear trend and exponential trend
- Our rolling-window estimator can be used (consistent and asymptotically normal)

Simulation-and-regression with parameter uncertainty

- We propose minimizing the Monte Carlo approximation to

$$R_{P_{\widehat{\mathrm{mix}}}}(f) = \mathbb{E}_{P_{\widehat{\mathrm{mix}}}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f(Y)]_t\right)\right] = \mathbb{E}_{\theta\sim\widehat{\mathrm{CD}}_n}\left[\mathbb{E}_{P_\theta}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f(Y)]_t\right)\right]\right]$$

for some loss L, where the expectation is w.r.t. the mixture distribution $P_{\widehat{\mathrm{mix}}} = \int_\Theta P_\theta d\widehat{\mathrm{CD}}_n(\theta)$ of sequences X, Y corresponding to some *estimated confidence distribution* $\widehat{\mathrm{CD}}_n$ over $\Theta \subset \mathbb{R}^p$ for θ_0

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  1. $N(\hat{\theta}, \hat{\Sigma}/n)$ where $\hat{\Sigma}$ is the estimate of the asymptotic covariance
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- Theory of confidence distributions from D. Liu, R. Y. Liu, and Xie (2022)

Connections to TSFMs and classical simulation-and-regression

- When $\widehat{\mathrm{CD}}_n$ is uniform over Θ, this is like TSFM pre-training (no information)

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- When $\widehat{\mathrm{CD}}_n$ is uniform over Θ, this is like TSFM pre-training (no information)
- When $\widehat{\mathrm{CD}}_n$ is Dirac delta at $\hat{\theta}$, this is like classical simulation-and-regression, which aims to minimize the Monte Carlo approximation to

$$R_{P_{\hat{\theta}}}(f) = \mathbb{E}_{P_{\hat{\theta}}}\left[\sum_{t \in \mathcal{T}} L(X_t - [f(Y)]_t)\right]$$

where expectation is w.r.t. distribution $P_{\hat{\theta}}$ of sequences X, Y corresponding to $\hat{\theta}$

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Questions we aim to answer

- **Calibration:** Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output multiple conditional quantiles?
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- **Improving fine-tuning and simulation-and-regression:** When does accounting for parameter uncertainty improve methods based on parameter point estimates?

Thank you!

Questions or comments?

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Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output conditional quantiles?

– For untrained transformers, see answer to question: "In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?"

– For pre-trained transformers, see answer to question: "When does a pre-trained transformer fine-tuned with our method outperform a transformer trained from scratch with our method?"

In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?

– When does this inequality hold:

$$\mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{P_{\widehat{\text{mix}}}}(Y)]_t\right)\right] < \mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{P_{\hat{\theta}}}(Y)]_t\right)\right]$$

where both expectations are with respect to the distribution P_{θ_0} of the sequences corresponding to the unknown true parameter θ_0, $f^*_{P_{\widehat{\text{mix}}}}$ is the minimizer in the function class of $R_{P_{\widehat{\text{mix}}}}(f)$, and $f^*_{P_{\hat{\theta}}}$ is the minimizer in the function class of $R_{P_{\hat{\theta}}}(f)$

When does a pre-trained transformer fine-tuned with our method outperform a transformer trained from scratch with our method?

– Intuitively, this occurs when the following conditions hold

– Large parameter uncertainty, e.g., when the sample size of the observed time series is small relative to the noise

– The distributions used in the pre-training phase are close to the true distribution, e.g., when we have domain knowledge or we have previously observed time series with similar distributions

What do we mean by "almost every" function? Prevalent subset

Definition (J. Yorke and Ott (2005))

Let V be a completely metrizable topological vector space. A Borel set $E \subset V$ is said to be **prevalent** if there exists some Borel measure μ on V such that:
1. $0 < \mu(C) < \infty$ for some compact subset C of V.
2. The set $E + v$ has full μ-measure (that is, the complement of $E + v$ has measure zero) for all $v \in V$.

– In this literature, it is common to say that, if $E \subset V$ contains a prevalent Borel set, then "almost every" element of V lies in E. In this sense, "almost every" smooth map from Θ to $\mathbb{R}^k$ has the following properties.

What is a smooth manifold? (University of Toronto Lecture Notes) I

– A C^∞-manifold is a Hausdorff topological space M, with countable basis, together with a C^∞-atlas.

– An n-dimensional manifold is a space that locally looks like $\mathbb{R}^n$. To give a precise meaning to this idea, our space first of all has to come equipped with some topology (so that the word "local" makes sense).

What is a smooth manifold? (University of Toronto Lecture Notes) II

– Recall that a *topological space* is a set M, together with a collection of subsets of M, called *open subsets*, satisfying the following three axioms: (i) the empty set $\emptyset$ and the space M itself are both open, (ii) the intersection of any finite collection of open subsets is open, (iii) the union of any collection of open subsets is open.

– The collection of open subsets of M is also called the topology of M. A map $f : M_1 \to M_2$ between topological spaces is called *continuous* if the pre-image of any open subset in M_2 is open in M_1. A continuous map with a continuous inverse is called a *homeomorphism*.

What is a smooth manifold? (University of Toronto Lecture Notes) III

– One ingredient in the definition of a manifold is that our topological space comes equipped with a covering by open sets which are homeomorphic to open subsets of $\mathbb{R}^n$.

– Let M be a topological space. An n-dimensional *chart* for M is a pair (U, ϕ) consisting of an open subset $U \subset M$ and a continuous map $\phi : U \to \mathbb{R}^n$ such that ϕ is a homeomorphism onto its image $\phi(U)$. Two such charts (U_α, ϕ_α), (U_β, ϕ_β) are C^∞-compatible if the transition map

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta)$$

is a diffeomorphism (a smooth map with smooth inverse). A covering $\mathcal{A} = (U_\alpha)_{\alpha \in A}$ of M by pairwise C^∞-compatible charts is called a C^∞-atlas.

– We recall that a topological space is called *Hausdorff* if any two points have disjoint open neighborhoods. A *basis* for a topological space M is a collection $\mathcal{B}$ of open subsets of M such that every open subset of M is a union of open subsets in the collection $\mathcal{B}$. For example, the collection of open balls $B_\varepsilon(x)$ in $\mathbb{R}^n$ define a basis. But one already has a basis if one takes only all balls $B_\varepsilon(x)$ with $x \in \mathbb{Q}^n$ and $\varepsilon \in \mathbb{Q}_{>0}$; this defines a countable basis. A topological space with countable basis is also called *second countable*.

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